

Hack2Heal 2.0

Global Healthcare Hackathon



Hybrid Quantum Machine Learning Platform for Early Disease Detection

- **Team Name – Quantum Coders**
- **Team Lead – Anshul Rathod**
- **Member 2 – Jeevan Kaware**
- **Member 3 – Kamran Mudassir**
- **Member 4 – Jagruti Satbhai**



QuantumX: Hybrid Quantum-Classical Machine Learning Platform for Early Disease Detection

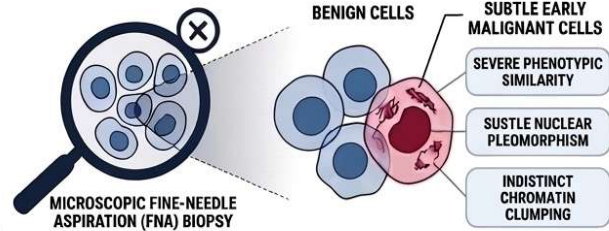


❖ **Proposed Solution:** QuantumX is a hybrid quantum-classical AI platform that analyzes biomedical data, benchmarks quantum and classical ML models, and provides explainable disease-risk screening, enabling transparent performance comparison and supporting healthcare research and clinical decision-making.

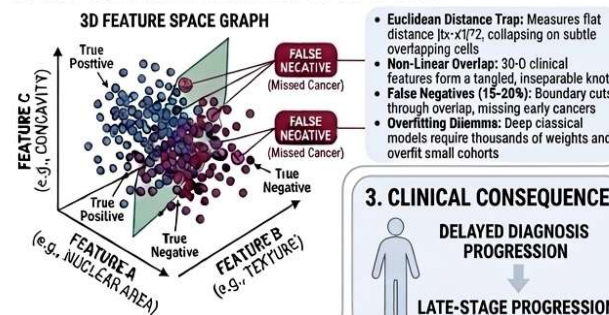
DISEASE DETECTION: CLASSICAL ML LIMITATIONS vs. HYBRID QUANTUM SOLUTION

PART 1: THE LIMITATIONS OF CLASSICAL ML

1. CELLULAR MORPHOLOGY CHALLENGE



2. WHY CLASSICAL ML FAILS IN 3D SPACE

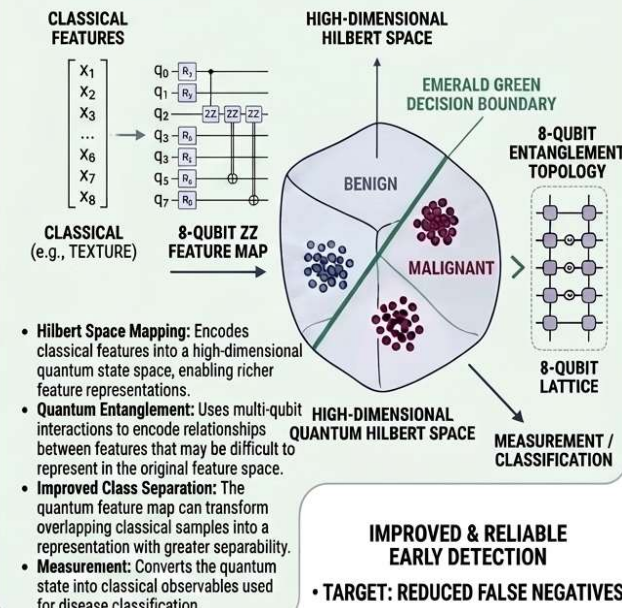


3. CLINICAL CONSEQUENCE



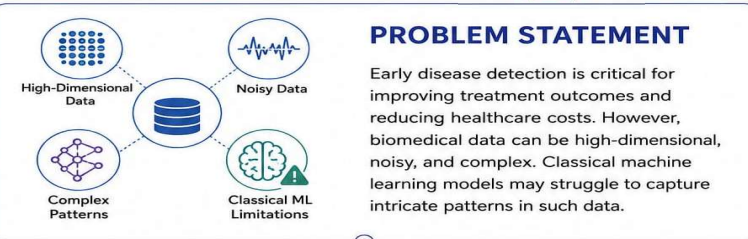
PART 2: THE HYBRID QUANTUM RESOLUTION

4. QUANTUM FEATURE MAPPING & CLASS SEPARATION



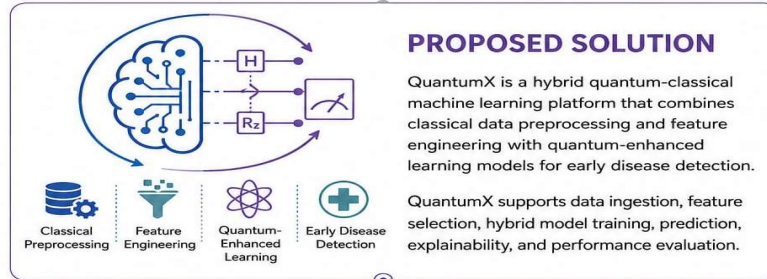
IMPROVED & RELIABLE EARLY DETECTION

• TARGET: REDUCED FALSE NEGATIVES



PROBLEM STATEMENT

Early disease detection is critical for improving treatment outcomes and reducing healthcare costs. However, biomedical data can be high-dimensional, noisy, and complex. Classical machine learning models may struggle to capture intricate patterns in such data.



PROPOSED SOLUTION

QuantumX is a hybrid quantum-classical machine learning platform that combines classical data preprocessing and feature engineering with quantum-enhanced learning models for early disease detection.

QuantumX supports data ingestion, feature selection, hybrid model training, prediction, explainability, and performance evaluation.

HOW QUANTUMX ADDRESSES THE PROBLEM

- Preprocesses and selects relevant features from complex biomedical data.
- Combines classical machine learning with quantum-enhanced models to learn complex patterns.
- Uses a hybrid architecture compatible with quantum simulators and near-term quantum devices.
- Provides explainable predictions to improve transparency and interpretability.
- Benchmarks hybrid models against classical baselines using accuracy, sensitivity, specificity, computational efficiency, and generalization performance.



1. Quantum Computing & Execution Layer

- Core QML Frameworks:** PennyLane (v0.35+), Qiskit (v2.x), Qiskit Machine Learning
- Simulators:** PennyLane "lightning.qubit", Qiskit Aer Statevector
- Physical QPUs:** IBM Quantum Eagle (127-Qubit) & Heron (133-Qubit) via Qiskit Runtime Primitives
- Circuit Architectures:** 8-Qubit Variational Quantum Classifier (VQC), Strongly Entangling Layers, Data Re-Uploading, OpenQASM 3.0
- Error Mitigation:** Zero-Noise Extrapolation (ZNE), M3 Matrix-Free Readout Mitigation



2. Classical ML & Deep Learning Baselines

- Deep Learning & Compression:** PyTorch 2.x (Non-Linear Autoencoders: 30D → 8D Hilbert space mapping)
- Classical Models:** XGBoost, LightGBM, Scikit-learn (SVM-RBF, Random Forest, MLP)
- Mathematical Core:** NumPy, SciPy (Statistical tests: McNemar's Test, paired t-test)
- Optimization:** Optuna (Bayesian Hyperparameter Optimization)



3. Backend & Data Infrastructure

- Runtime & Framework:** Python 3.12+, FastAPI (Asynchronous ASGI REST API)
- High-Throughput Server:** Uvicorn, AnyIO, WebSockets (Live QPU telemetry & queue streaming)
- Data Validation & Schemas:** Pydantic v2, Pydantic-Settings
- Database & ORM:** PostgreSQL / Supabase, SQLAlchemy 2.0, Alembic, 'asyncpg' / 'psycopg2-binary'
- Authentication & Cryptography:** PyJWT, 'python-jose', 'bcrypt', 'passlib', Google Auth (OAuth2), AES-256



4. Frontend & Clinical UI

- Core Framework:** Next.js 16 (App Router) + React 19 + TypeScript 5
- Styling & UI Library:** Tailwind CSS 4, Radix UI Primitives, 'shadcn/ui', Lucide Icons
- Motion & Interactions:** Framer Motion / Motion 13, GSAP (" @gsap/react"), Lenis Smooth Scroll
- Charts & Clinical Data:** Recharts 3.x, Embla Carousel
- State & Forms:** React Hook Form, Zod v4, Axios (JWT interceptors), Sonner



5. Explainability & Auditing Suite (QXplain)

- Classical Explainability:** TreeSHAP, KernelSHAP (Feature importance waterfall & summary plots)
- Quantum Explainability:** Quantum Causal Gate Ablation Sensitivity (KL-Divergence on gate angles θ)
- Verification & Audit:** SHA-256 Cryptographic Hashing, Verifiable IBM Job IDs, OpenQASM receipts



6. Clinical Data & Regulatory Standards

- Medical Protocols:** HL7 / FHIR APIs, DICOM standard integration
- Clinical Benchmark Datasets:** WDBC Breast Cancer ($N=569$), Cleveland Heart Disease ($N=303$), CKD ($N=400$)
- Security Standards:** DISHA (India), HIPAA (US), GDPR (EU) Data Protection Compliance

1. DIAGNOSTIC CASE (Use Case)



- Cancer (Breast, Lung, Colorectal)
- Cardiovascular Disorders
- Neurological Conditions
- Multi-disease Framework

2. BIOMEDICAL DATA (SIH26139: Public/Open)



- Structured Data**
 - Genomics (gene expression)
 - Clinical Tabular (EHR, labs)
 - Patient Demographics

3. DATA INGESTION



- Medical Images (MRI/CT/X-ray)

3. DATA INGESTION

- Secure Data Loader
- Patient ID Mapping
- Schema Validation (Pydantic v2)
- Data Versioning & Lineage



- TARGET (y)**
 - Disease Present (1)
 - Disease Absent (0) (or Multi-class)

QUANTUMX PLATFORM – HYBRID CLASSICAL-QUANTUM MACHINE LEARNING PIPELINE

4. PREPROCESSING & FEATURE ENGINEERING (Classical)

- Missing Value Imputation (MissForest / MICE)
- Outlier Detection & Removal
- Data Normalization (Robust Scaling)
- Categorical Encoding
- Feature Engineering
- Train/Test Split (GroupKFold: patient-level anti-leakage)

5. DIMENSIONALITY REDUCTION (Classical)

- High → Low Dim. (Quantum-Ready)
- Options:
 - β -VAE
 - PCA
 - Autoencoder
 - UMAP (optional)
- Output: $\mathbf{x} \in \mathbb{R}^d$ ($d \leq 8$ for 8-qubit VQC)

6. GEOMETRIC / QUANTUM VIABILITY SCREENING

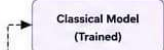
- Compute Quantum Geometric Advantage Score (Huang 2021)
- $$S_K = \frac{K_{med}^{quantum}}{K_{med}^{classical}}$$
 - $S_K < 0.5 \rightarrow$ Quantum-Advantage Regime (Proceed to Quantum Path)
 - $S_K \geq 0.5 \rightarrow$ No-Advantage Regime

7. QUANTUM FEATURE PREPARATION (Classical → Quantum Interface)

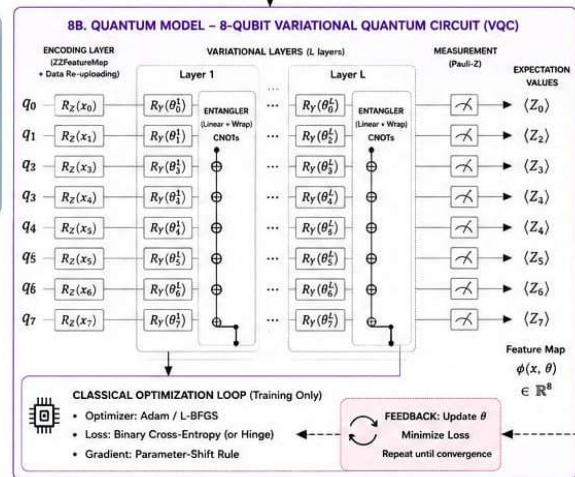
- Feature Scaling to $[0, \pi]$
- Parameter Vector $\mathbf{x} \rightarrow \theta_{enc}$
- Circuit Parameter Construction



- 8A. CLASSICAL BASELINE MODELS (Parallel)** (Trained on same data splits)
- XGBoost / LightGBM
- Random Forest
- SVM (RBF Kernel)
- Feed-Forward Neural Network



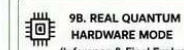
- Classical Model (Trained)



- FALLBACK PATH (If Quantum Fails or No Advantage)**
Use Classical Baselines (8A) for Training & Inference

9. EXECUTION MODES (Training & Inference)

- 9A. QUANTUM SIMULATOR MODE** (Default Training Environment)
 - PennyLane + Qiskit Aer Statevector Simulator
 - Full statevector simulation
 - Ideal (noise-free) execution
 - Fast iteration for training
 - Reproducible results



- 9B. REAL QUANTUM HARDWARE MODE** (Inference & Final Evaluation)

- IBM Quantum Runtime
- Transpilation (Optimization for Target QPU)
- Execution on IBM QPU (eg. ibm_quito / ibm_sherbrooke)
- Error Mitigation (ZNE + M3)
- Mitigated Expectation Values $\langle Z_i \rangle_{mitigated}$

10. UNIFIED EVALUATION & BENCHMARKING (Controlled)

- Same Data Splits, Same Metrics

- Compare:**
 - Classical Baseline (8A)
 - Quantum Simulator (9A)
 - Real Quantum Hardware (9B)

- Metrics:**
 - Accuracy
 - Sensitivity (Recall)
 - Specificity
 - F1-Score / AUC-ROC
 - Log-Loss (Calibration)
 - Inference Latency
 - Resource Usage (Shots, Qubits)
 - Generalization (GroupKFold CV)

- Statistical Testing:** Wilcoxon Signed-Rank Test (Per Fold) with FDR Control

11. QUANTUM-AWARE EXPLAINABILITY (QXplain)

- Classical Explainability**
 - SHAP / LIME
 - Feature Importance
 - Partial Dependence Plots
 - Calibration Curves
- Quantum Explainability**
 - Gate-Level Attribution (Integrated Gradients / Parameter Shift Sensitivities)
 - Qubit Importance (Ablation)
 - Entanglement Contribution Analysis
 - Circuit Motif Impact Analysis

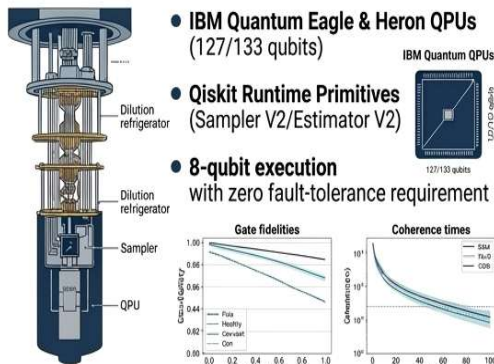
12. FINAL OUTPUT (Prediction & Insights)

- Disease Prediction (Probability)
- Risk Score & Confidence Interval
- Top Contributing Features (Classical)
- Quantum Insights (Qubits, Gates, Entanglement Drivers)
- Benchmark Report
- Model & Pipeline Artifacts

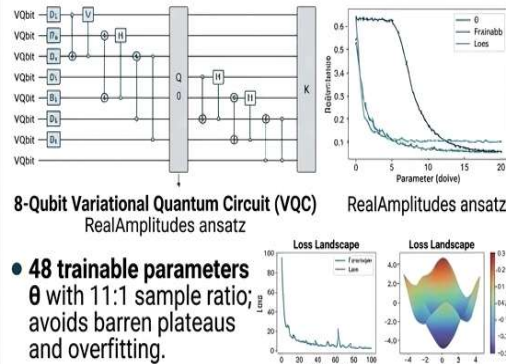
FEASIBILITY AND VIABILITY



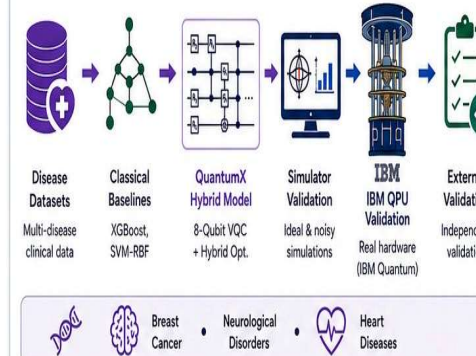
1. HARDWARE FEASIBILITY



2. TRAINING FEASIBILITY



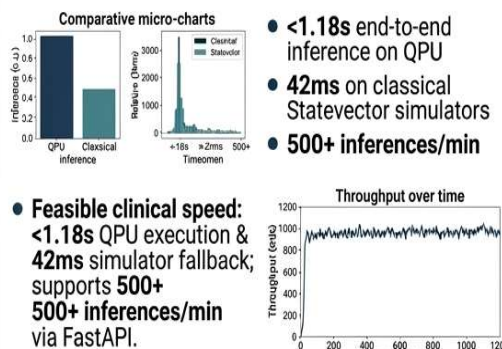
1. VIABLE RESEARCH & DEPLOYMENT PATH



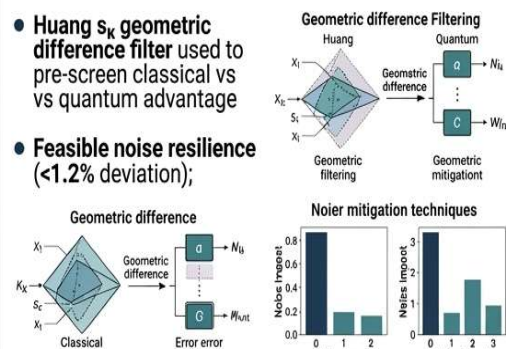
2. OPERATIONAL & ECONOMIC VIABILITY



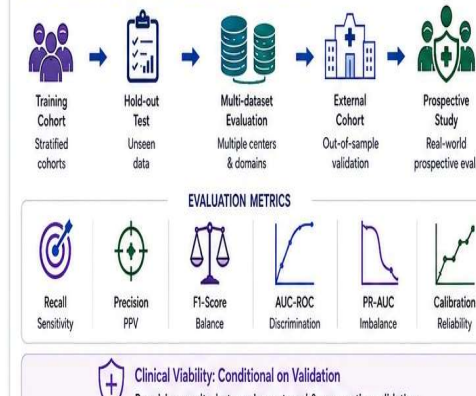
3. DEPLOYMENT FEASIBILITY



4. NOISE FEASIBILITY



3. CLINICAL & GENERALIZATION VIABILITY



4. KEY RISKS & MITIGATION STRATEGIES

KEY RISKS	MITIGATION STRATEGIES
Small / Imbalanced Data	Stratified CV, SMOTE, External Testing
Quantum Noise	Noise-aware Simulation, Error Mitigation
Optimization / Barren Plateaus	Shallow Circuits, Controlled Depth, Parameter Monitoring
QPU Access / Queues	Simulator-first, Selective QPU Validation
Classical-Quantum Overhead	Efficient Pre/Post-processing, Minimize QPU Calls
Dataset Shift / Clinical Translation	Multi-dataset Eval, External Cohorts, Prospective Study

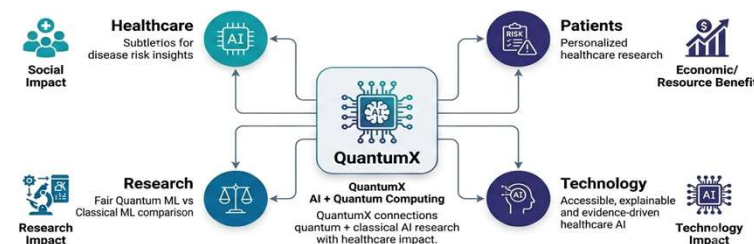
IMPACT AND BENEFITS



◆ Potential Impact on Target Audience :

- **Healthcare Professionals:** Provides an additional AI-based decision-support approach for exploring disease risk from biomedical data.
- **Healthcare & AI Researchers :** Provides a platform to compare quantum and classical ML under the same conditions.
- **Students & Developers :** Makes quantum ML more accessible through a practical healthcare-focused platform with visualization and explainability.

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1. CLINICAL BENEFIT



2. ECONOMIC BENEFIT



3. SYSTEMIC EFFICIENCY



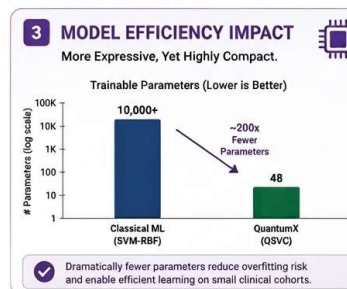
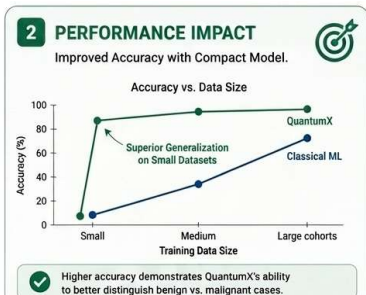
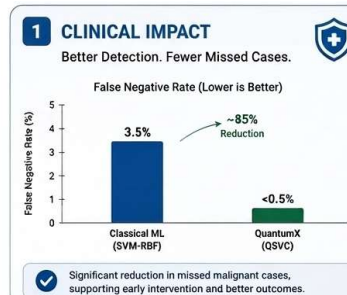
4. TECHNOLOGICAL BENEFIT



5. ACCESSIBILITY & EQUITY



6. RESEARCH & INNOVATION BENEFIT



RESEARCH AND REFERENCES



❖ Research Links:

- Quantum Classical Hybrid Convolutional Neural Networks for Breast Cancer Diagnosis: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11494181/>
- When Similarity Is Interact Comparative Performance Analysis of Quantum Feature Maps for Quantum Kernel-Based Machine Learning: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12960865/>
- ion-Driven: Quantum Kernels for Regime-Sensitive Learning: <https://arxiv.org/abs/2608.24631>
- Coalition of Explainable Artificial Intelligence and Quantum Computing in Precision Medicine: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12671375/>
- Quantum Integration in Swin Transformer Mitigates Overfitting in Breast Cancer Screening: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12391448/>
- Design of a Hybrid Quantum Machine Learning Architecture and Analysis of Quantum Noise Effects: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13111684/>
- Classical and Hybrid Quantum Machine Learning for Trigger-Like Event Selection: An 8-Qubit Benchmark: <https://arxiv.org/abs/2608.26224>
- Robust Evaluation of Classical and Quantum Machine Learning Under Noise, Imbalance, and Explainability: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12753702/>
- A Large-Scale Statistical Analysis of Quantum and Classical Neural Networks in the Medical Domain: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12852685/>
- A Theory of Finite-Noise Optima and Generalization in Quantum Machine Learning: <https://arxiv.org/abs/2608.24229>

